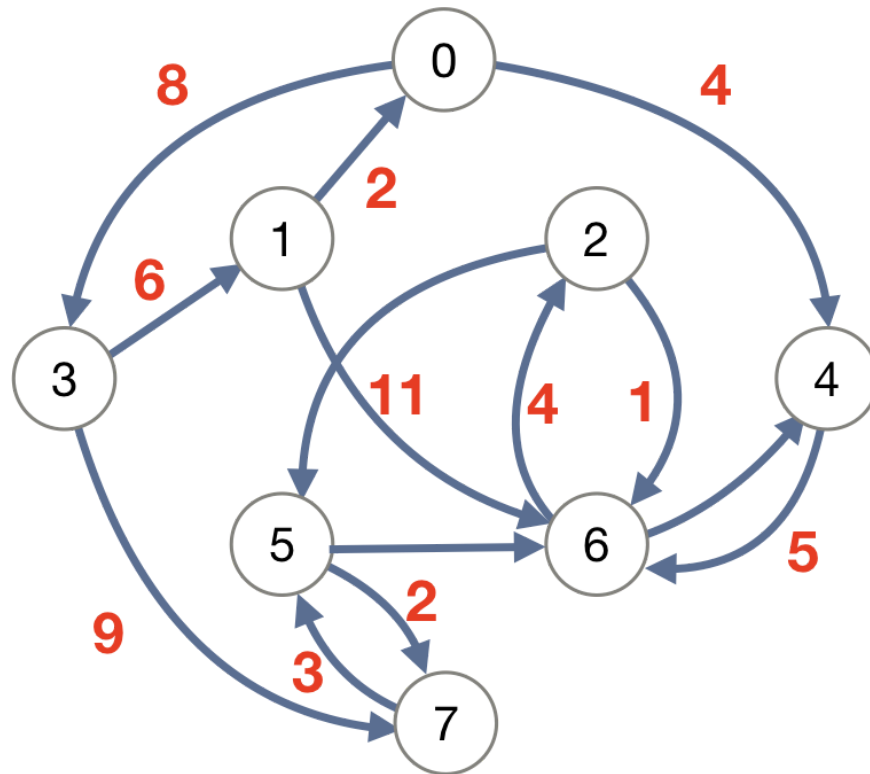


Practice Problems for CSE 2202 Lab Final



Graph Representations:

1. Represent the above graph in -
 - a. Adjacency Matrix
 - b. Adjacency List
 - c. Edge List

Graph Traversal:

2. Determine the traversal sequence for the above graph:
 - a. Breadth-First Search
 - b. Depth-First Search

*Minimum Spanning Tree:

3. Determine the minimum cost to visit all vertices of the graph
 - a. Prim's algorithm
 - b. Kruskal's algorithm

*Single-Source Shortest Path:

4. Determine the shortest path starting from vertex 0 of the graph
 - a. Dijkstra's algorithm (for non-negative edges)
 - b. Bellman-Ford's algorithm (for negative edges)

All-Pair Shortest Path:

5. Determine the shortest path between all pairs of vertices of the graph
 - Floyd-Warshall's algorithm

*n-th Fibonacci:

6. Given n, determine the n-th Fibonacci number using -
 - a. Top-down approach / Memoization
 - b. Bottom-up approach / Tabulation

*Longest Common Subsequence:

7. For the following two strings,

s1 = "bd"

s2 = "abcd"

Determine the LCS length using -

- a. Top-down approach / Memoization
- b. Bottom-up approach / Tabulation

0/1 Knapsack:

8. Given,

Knapsack capacity, $W = 40$

Item values = {10, 20, 30, 40}

Item weights = {30, 10, 40, 20}

Determine the maximum attainable value for the given items within the capacity.